

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 226 — Past Questions & Materials
by Renance

Course: EMT 226
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 7
Levels: 200

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Paper 1 of 7 — EMT 226 Introductory Climatology and Biogeography 2023/2024

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

SECOND SEMESTER 2023/2024 DEGREE EXAMINATION

EMT226: INTRODUCTORY CLIMATOLOGY AND BIOGEOGRAPHY (2 UNITS)

TIME ALLOWED: 3HRS

INSTRUCTION: ANSWER FOUR QUESTIONS

ef Write briefly on the following types of Biogeography

a. Ecological Biogeography __b. Island Biogeography c. historical
Biogeography d. Landscape Biogeography e. Conservation Biogeography

vb Discuss in details any five Climatological problems

__b__ Explain any two-of the problems——

3. Describe the relationship between climate and vegetation.

_ 4. Clarify the relationship between-soil-type-and vegetation-

"Write short note five Climate Data Analysis Tools

6A Explain five climatic factors that influence temperature

Paper 2 of 7 — EMT 226 Introductory Climatology and biogeography 2024/2025

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

SECOND SEMESTER 2022/2023 DEGREE EXAMINATION

EMT226: INTRODUCTORY CLIMATOLOGY AND BIOGEOGRAPHY
(2 UNITS)

TIME ALLOWED: 3HRS

INSTRUCTION: ANSWER FOUR QUESTIONS. TWO FROM EACH SECTION

SECTION A

1 Define the following terms:

- a. Physical climatology
- b. Regional climatology
- c. Applied climatology
- d. Marine climatology *bio-geography*
- e. Island climatology *no-geography*

☒ 2a. List any five of climatological problems

b. Explain any two of the problems

☒ 3a. Mention only five elements of climate

b. Discuss any two of the elements of climate

temp

SECTION B

☒ 4a. Define the term Temperature

4b. What is the difference between climate and weather?

5 Write short note on the following types of rainfall

- i. Conventional
- ii Relief
- iii. Cyclonic

☒ 6. Explain how the following elements affect temperature:

i. Sunshine ✓

ii. Latitude ✓

iii. Relief ✓

iv. Ocean Currents

v. Altitude ✓

Relief refers to the overall shapes of lands and their surface like valley, ridges

Paper 3 of 7 — EMT 226, 2024 INTRODUCTION CLIMATOLOGY AND BIOGEOGRAPHY past question

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
SECOND SEMESTER 2022/2023 DEGREE EXAMINATION
EMT226: INTRODUCTORY CLIMATOLOGY AND BIOGEOGRAPHY
(2 UNITS)
TIME ALLOWED: 3HRS
INSTRUCTION: ANSWER FOUR QUESTIONS. TWO FROM EACH SECTION
SECTION A

1 Define the following terms:

- Physical climatology
- Regional climatology
- Applied climatology
- Marine climatology *bio-geography*
- Island climatology *eco-geography*

2a. List any five of climatological problems

b. Explain any two of the problems

3a. Mention only five elements of climate

b. Discuss any two of the elements of climate

SECTION B

4a. Define the term Temperature

4b. What is the difference between climate and weather?

5 Write short note on the following types of rainfall

- Conventional
- Relief
- Cyclonic

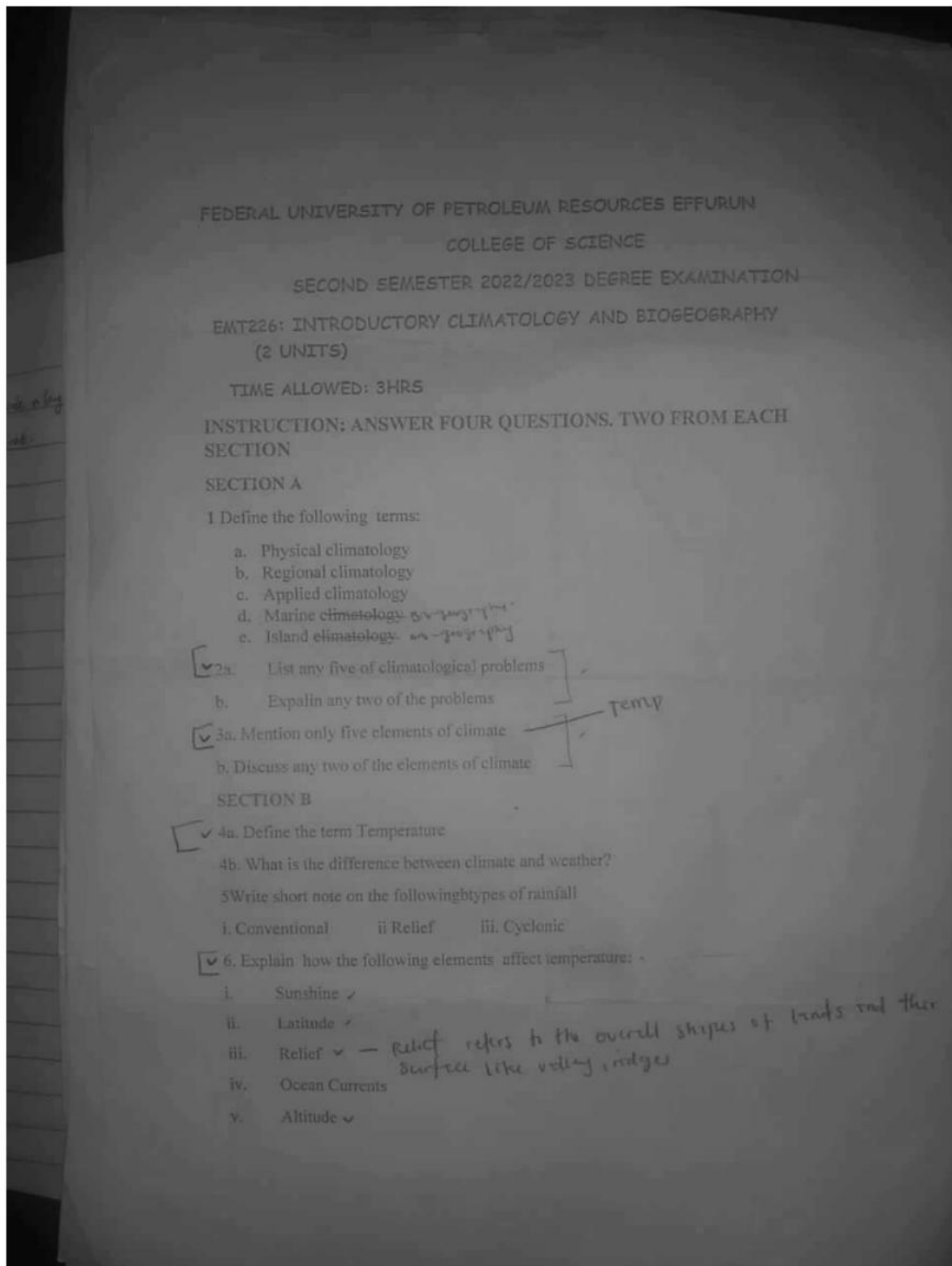
6. Explain how the following elements affect temperature:

- Sunshine ✓
- Latitude ✓
- Relief ✓ — *relief refers to the overall shapes of lands surface like valley, ridges*
- Ocean Currents
- Altitude ✓

Paper 4 of 7 — EMT 226

Level 200 • Second Semester • PastQuestion

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Paper 5 of 7 — LECTURE MATERIAL ON EMT 226 INTRODUCTORY CLIMATOLOGY AND BIOGEOGRAPHY

Level 200 • Second Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM

RESOURCES

DEPARTMENT OF ENVIRONMENTAL

MANAGEMENT AND TOXICOLOGY

LECTURE MATERIAL ON EMT 226 INTRODUCTORY

CLIMATOLOGY AND BIOGEOGRAPHY

Climate Data Analysis Tools and

Methods

Overview: Climate Data Processing

The first rule of data processing is to look at your data!

What data format is being used? What variables are available and what are the units? What type of grid: rectilinear, curvilinear or unstructured?

What is the orientation of the grid: north-to-south or south-to-north?

What longitude order? What temporal and vertical coordinate(s) are used? Are there missing values present? (If so, how are they identified?)

Does the file contents adhere to any convention? Do not make the assumption that the data are without problems. Unfortunately, processing climate data can be rather intimidating due the large sizes of the data sets. Some climate model components (eg, ocean) can produce individual monthly mean files that exceed 17GB while some observationally based reanalysis datasets exceed several tera bytes .

Since climate, by definition, is the statistics of one or more variables over a period of time, it often means that there are many files to be processed.

Further, some climate projection scenarios may span 50-100 years into the future and there may be multiple model runs (ensembles) to assess

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model sensitivity. Assorted data formats and a variety of processing tools may be used by different groups to archive and process the data.

1. Statistical & Diagnostic Methods Overview

By definition, climate is the statistics of weather over an arbitrarily defined time span. The methods used to derive the statistical estimates can be simple or very complex. The most common statistic is the average of some variable (eg., temperature). However, solely focusing on the

average can be misleading. For example, the average temperature may be consistent with previous time spans but the variance may have changed in some 'significant' way. The gradual change of some variable over time is defined as a trend. Extreme value analysis (EVA) is used to assess the likelihood of events at the tails of a variable's distribution. Classical statistical methods use the assumption of stationarity which implies that a variable's distribution (e.g., mean, variance, no trend, etc.) does not vary with time. Obviously, the stationarity assumption is violated under a climate changing (eg., warming) scenario. However, there are preprocessing approaches such as detrending which allow the resulting data to be treated as 'approximately' stationary.

2. Binary

Binary files encountered in climate are generally created from compiled languages such as fortran or C/C++. These are easy to create and are supported by the compiled languages. Some software tools used in climate research (IDL, Matlab) can produce binary files but they are proprietary and are best read with the tools that produce them. Binary

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files created by fortran/C/C++ require that the user know, a priori, the following:

- (a) The structure or lay-out of the data;
- (b) The data types [eg., integer, float, double] of all the variables;
- (c) The "endian" type of the file being read (big endian or little endian);
- (d) Whether the file is a 'flat' binary file or if it was written with, say, fortran a sequential write statement, in which case, there are hidden record separators.

Without this external collection of information, it is difficult to unambiguously ascertain a binary file's contents. As a result there are no generic file tools to deal with binary files. This is one reason why binary files are not in general use for archiving climate data.

3. Climate Data Processing Software

There is no simple answer. All software tools and languages have strengths and weaknesses. For large scale data processing on a variety of data sets in assorted data formats and differing project requirements, it

is unlikely that a perfect tool or language exists. Often, a combination of software tools and languages will be needed. Climate data processing involves 3 components: (1) file handling (I/O); (2) processing (data manipulation and computations), and (3) graphics (visualization). There are three different software categories used for climate data processing

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and visualization: (1) compiled languages (eg., fortran, C, C++); (2) command line operators and viewers (NCO, CDO, ncview, panoply); (3) interpreted languages (NCL, GrADS, Ferret, R, Generic Mapping Tools (GMT), Perl Data Language (PDL) , Python [CDAT/PyNIO/PyNGL/Numpy/matplotlib], and the commercial products Matlab, IDL and, to a lesser extent, PV-Wave).

Compiled languages can be much faster than the interpreted languages for large computation bound tasks. Language compilers analyze and optimize code and create machine specific execution instructions. As a result, they can perform looping (iterations) faster. For example, weather forecast and climate models are often written in fortran (usually f90). However, compiled languages lack builtin support for accessing the different data formats used in climate studies and they have no builtin graphics. Further, programming in compiled languages can be tedious.

Command line operators (CLOs) are tools that can be executed directly at the system prompt line. There are many NCO and CDO operators and there is some functional overlap. Each operator is designed to perform a specific task efficiently. For example, the NCO operator "ncra" can input one or more netCDF files, compute time averages (means) of all or selected variables on the file(s) and save the results to a netCDF file. It is not uncommon to use an NCO/CDO operator to accomplish a specific task and, then, feed the output file to a different CDO/NCO operator. Ncview is a commonly used visual browser for netCDF format files.

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Interpreted languages are general purpose software tools. They include support to read and write assorted data formats; have many builtin computational functions; and, create visualizations. These tools have all the capabilities of the CLOs and ncview and can do much more.

However, they do require users to enter commands interactively or via a script.

Within NCAR's Climate Analysis Section, tera-bytes of model output, observationally based data sets like the reanalysis products (ERA-Interim, MERRA, NCEP-NCAR, JRA, ...) and satellite data are analyzed, evaluated and used as the basis for publications. The data are in a variety of formats, including: netCDF-3/4, GRIB-1/2, HDF4, HDF4-EOS, HDF5, HDF5-EOS. The primary post-processing tools used are NCL and the NCO. In some cases, data created by NCL/NCO are input to R for certain statistical methods not available within NCL. Depending upon the application, the CDO, IDL and Matlab are also used.

4. Common Climate Data Formats

Data formats commonly encountered in climate research fall into 3 generic categories: GRIB, netCDF and HDF. All of these formats are portable (machine independent) and self-describing. Self-describing files can be examined and read by the appropriate software without the user knowing the file's structural details. Further, additional information about the data, called "metadata", may be included in the file. Typical metadata may include textual information about each variable's contents

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and units (eg., "specific humidity" and "g/kg") or numerical information describing the coordinates (eg., time, level, latitude, longitude) that apply to the variables on the file.

Each of these file formats has evolved over time to address the changing needs of the communities they support. Hence, there are multiple versions of each format. Unfortunately for users, the newer formats are not necessarily backward compatible despite the similar naming conventions. This can be both frustrating and confusing for users.

Specifically, the data formats are as follows:

GRIB1: GRIdded Binary (Edition 1), World Meteorological Organization

GRIB2: GRIdded Binary (Edition 2), World Meteorological Organization

netCDF3: Network Common Data Form, (Version 3.x), Unidata
(UCAR/NCAR)

netCDF4: Network Common Data Format, (Version 4.x), Unidata
(UCAR/NCAR)

HDF4: Hierarchical Data Format, (Version 4.x), NCSA/NASA

HDF4-EOS2: HDF4-Earth Observing System, (Version 2; georeferenced data)

HDF5: Hierarchical Data Format, (Version 5.x), NCSA/NASA

HDF5-EOS5: HDF5-Earth Observing System, (Version 5; georeferenced data)

GeoTIFF: Georeferenced raster imagery

A subtle difference is that netCDF/HDF/HDF-EOS are file formats while GRIB is a record format. Because netCDF/HDF/HDF-EOS are file formats there are rules on a file's contents. For example, a simple netCDF rule is that all variable names must be unique. HDF allows a file to contain

6 multiple variables with the same name BUT the variables must be in different 'groups'. These 'groups' can be complicated but for this gross overview they can be considered analogous to Unix directories (Windows folders). Each GRIB-1 record (aka, 'message') contains information for two horizontal dimensions (eg., latitude and longitude) for one time and one level. GRIB-2 allows each record to contain multiple grids and levels for each time. A collection of GRIB records is called a GRIB file.

However, there are no rules dictating the order of the collection of GRIB records (eg, records can be in random chronological order).

5. Empirical Orthogonal Function (EOF) Analysis and Rotated EOF Analysis

In climate studies, EOF analysis is often used to study possible spatial modes (ie, patterns) of variability and how they change with time (e.g., the North Atlantic Oscillation). In statistics, EOF analysis is known as Principal Component Analysis (PCA). As such, EOF analysis is sometimes classified as a multivariate statistical technique. However, there is no a priori hypothesis based on some probability distribution and, hence, no statistical test. Further, EOF analysis is not based on physical principles. Rather, a field is partitioned into mathematically orthogonal (independent) modes which sometimes may be interpreted as atmospheric and oceanographic modes ('structures'). Typically, the EOFs are found by computing the eigenvalues and eigenvectors of a spatially

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weighted anomaly covariance matrix of a field. Most commonly, the spatial weights are the $\cos(\text{latitude})$ or, better for EOF analysis, the $\sqrt{\cos(\text{latitude})}$. The derived eigenvalues provide a measure of the percent variance explained by each mode. Unfortunately, the eigenvalues are not necessarily distinct due to sampling issues. North et al (Mon. Wea. Rev., 1982, eqns 24-26) provide a 'rule of thumb' for determining if a particular eigenvalue (mode) is distinct from its nearest neighbor. Atmospheric and oceanographic processes are typically 'red' which means that most of the variance (power) is contained within the first few modes. The time series of each mode (aka, principle components) are determined by projecting the derived eigenvectors onto the spatially weighted anomalies. This will result in the amplitude of each mode over the period of record.

By construction, the EOF patterns and the principal components are independent. Two factors inhibit physical interpretation of EOFs: (1) the orthogonality constraint and (2) the derived patterns may be domain dependent. Physical systems are not necessarily orthogonal and if the patterns depend on the region used they may not exist if the the domain changes. Still, even with these short comings, classical EOF (PCA) analysis has proved to be useful.

Rotated EOF analysis (REOF) : To address some of the limitations of classical EOF analysis, researchers 'rotate' the classical EOFs. Some rotational methods retain the orthogonality of the modes but not the principal components or vice versa. Oblique rotations preserved neither.

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Most commonly, the varimax rotation has been used. The objective is to minimize the mode complexity by making the large loadings larger and the small loadings smaller. Initially, a standard EOF analysis is performed and an EOF subset (eg, 10) is retained and subjected to varimax rotation. Some feel that the resulting patterns are more physically interpretable. Still, REOF methods have issues. The patterns are may still be domain dependent and the initial number of EOFs retained is arbitrary.

6. GRIB

The World Meteorological Organization (WMO) Commission for Basic

Systems in 1985 approved a general purpose, bit-oriented data exchange format: GRIB (GRIdded Binary). It is an efficient vehicle for transmitting large volumes of gridded data to automated centers over high-speed telecommunication lines using modern protocols. By packing information into the GRIB code, messages (or records - the terms are synonymous in this context) can be made more compact than character oriented bulletins, which will produce faster computer-to-computer transmissions. GRIB can equally well serve as a data storage format, generating the same efficiencies relative to information storage and retrieval devices." There are two GRIB versions denoted here as GRIB-1 (original) and GRIB-2 (new). Each GRIB-1 record contains information for two horizontal dimensions (eg., latitude and longitude) for one time and one level. GRIB-2 allows each record to contain multiple horizontal grids and levels for each time. A collection of GRIB records is called a GRIB file.

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However, there are no rules dictating the order of the collection of records.

Depending upon one's point of view, GRIB may or may not be described as "self describing." All the information necessary to unpack the data is contained within each record. However, the variable being unpacked is denoted by an "indicator parameter" and the variable's name and units must be determined by a table lookup.

Generally, operational weather centers create reanalysis and forecast products in GRIB format.

7. HDF

The Hierarchical Data Format is available in two versions: the original HDF4 and the more recent HDF5. Unfortunately, HDF4 and HDF5 interfaces and data models are completely incompatible. The HDF5 data model is more flexible and is a "a true hierarchical file structure, similar to the Unix file system." HDF5 does have some new features that are appealing to climate research, such as, parallel I/O and variable compression. (Note: Our experience is that any compression level greater than '1' is not worth the additional time required for the increased compression.)

The HDF{4/5}-EOS format extension defines three additional data types

based on HDF objects: grid, point, and swath. These data types allow the file contents to be referenced to Earth coordinates, such as latitude and longitude, and to time. An irritating issue is that standard HDF library
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calls cannot readily access geolocation or time data or metadata. Users must use the HDF{4/5}-EOS interfaces. The converse is also true. Certain information can only be accessed via standard HDF4 interfaces. Hence, both interfaces may be required to get at all the information on the file.

8. netCDF Overview

NetCDF (Network Common Data Form) is designed to facilitate access to array-oriented scientific data. NetCDF is a portable, "self-describing" format. This means that there is a header which describes the layout of the rest of the file, in particular the data arrays, as well as arbitrary file metadata in the form of name/value attributes. The additional information about a file or variable is commonly called "meta data" (information about the data). In some cases, the file's contents were written using a standard netCDF convention which ensures users and automatic software that certain 'rules' have been followed when creating the file. The two most commonly used netCDF conventions for climate data are the COARDS and CF conventions. The COARDS convention, created in 1995, is simple and easy to read. The CF convention was created in 2003 to address the evolution of models and data sets. It is a superset of the COARDS convention. It is quite precise in language. Unfortunately, the CF documentation can read like a legal document. A netCDF file can be checked for CF compliance by clicking the NCAS-CMS link: [cf-checker](#).

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Many commercial and public software tools support netCDF. Further, there are APIs (Application Programmer Interfaces) for a number of compiled languages like fortran, C and C++.

netCDF is the format most commonly used for climate model generated data. The following is a sample 'dump' of a typical netCDF file created by a climate model. It has several components: (a) dimension names; (b) dimension sizes of the dimension names; (b) the variables on the file which often include additional information about each variable and

temporal/spatial coordinates; and (c) global attributes which contain information about the file's contents. This sample file contains coordinate variables (CVs) which are defined as one-dimensional variables with the same name as a dimension. CVs should not have any missing data (for example, no `_FillValue` or `missing_value` attributes) and must be strictly monotonic (values increasing or decreasing). In the following, `time(time)`, `lat(lat)`, `lon(lon)` and `lev(lev)` are classified as coordinate variables while `PS` and `T` are classified as variables.

The UNLIMITED dimension size is used when a variable can grow to any length along that dimension. The unlimited dimension index is like a record number in conventional record-oriented files. Some software tools (eg, the netCDF Operators) require an unlimited dimension to perform certain operations. It is recommended that netCDF files be created with the unlimited dimension.

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There are two versions of netCDF- $\{3,4\}$. The netCDF-3 data model was used for many years and is often referred to as netCDF-classic. However, as datasets became larger, the grids more complicated, and user desires for more flexibility developed, the limitations of the netCDF-3 data model became apparent. For example, netCDF-3 does not support compression, string variables or parallel processing. To address these issues, netCDF-4 (nc4) was created. An nc4 file is a hybrid: a subset of HDF5 with netCDF-3 style API interfaces used to create and access the data. Hence, a nc4 file is actually a HDF-5 file 'under-the-hood.' Many of the commonly used post-processing tools can readily handle netCDF-4 files.

9. Regridding

Regridding is the process of interpolating from one grid resolution to a different grid resolution. This could involve temporal, vertical or spatial ('horizontal') interpolations. However, most commonly, regridding refers to spatial interpolation. There are numerous grid interpolation methods and users should choose the method appropriate for the intended task. Using an inappropriate interpolation scheme may lead to misleading results. Further, most climate grids are georeferenced on a sphere where pole singularities and the convergence of the longitude meridians can be issues that need to be addressed.

Grids used in climate research fall into several categories: regular, rectilinear, curvilinear and unstructured. For our purposes, grids described by one-dimensional latitude and longitude coordinates ([*]) will

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be called rectilinear; grids where the coordinates are two-dimensional ([*][*]) will be called curvilinear; and, grids in which the grid coordinates require a list of nodes (connectivity information) will be called unstructured. Why the variety of grids? Reasons include advances in computer capabilities, computational efficiency, addressing pole singularities and physical constraints.

Quantitative evaluation of data contained on different grids requires regridding to a common grid. There are many regridding methods. Classic interpolation methods include: bilinear, nearest neighbor, inverse distance, spline, binning, spectral and triangulation. Certain applications of regridded data may necessitate two specialized interpolation methods to achieve required physical (eg, flux conservation) or mathematical (eg, higher order derivatives) requirements: conservative and patch.

The most commonly used methods of climate grid interpolation are bilinear, conservative and patch. The bilinear method is easy to program and apply when the source and destination grids are rectilinear. Virtually all data processing tools provide a function to implement some form of bilinear interpolation for rectilinear grids. However, curvilinear and unstructured grids may involve sophisticated searching algorithms to determine the points surrounding the location to be interpolated.

Some interpolation schemes use a multi-step approach. Initially, the user specifies the type of interpolation to be performed (eg, bilinear,

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conservative or patch) and the source and destination grid spatial coordinates. The software then generates a netCDF file which contains an array of weights. Depending upon the grid structures and the sizes, this may take a significant amount of time due to searching requirements. Finally, the user can read the generated weights and perform a 'sparse matrix multiply' to effect the regridding. This is a very fast operation and is ideally suited to processing a large number of grids.

Which Spatial Interpolation Method Should be used?

The choice of which method to use should be guided by a combination of the intended use of the interpolated data and the structure of the original variable. Most geophysical variables (eg., temperature) are highly spatially correlated at all time scales. Some variables (eg., precipitation) may have little or no spatial correlation at high frequency sampling (eg., 3-hourly) but on monthly, seasonal or annual mean time scales may be 'reasonably' smooth in space. In other instances, there may be physical requirements (e.g., conservation of energy) that require a specific interpolation method. There are data for which no interpolation method should be used. An example would be categorical data for land (eg., desert, rain forest, ...).

For smoothly varying variables, bilinear interpolation is adequate for many applications. For discontinuous variables (eg, 3 hourly precipitation), bilinear interpolation may not be appropriate. For

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example, consider a 3-hourly precipitation data set at 0.25 degree resolution (eg., TRMM) that is to be interpolated to a 2 degree grid. At the equator, within each 2 degree grid there are 64 0.25 grid points. Many grid points may have no precipitation (0.0) while a few have large values (eg., local convective activity). If bilinear interpolation was used, it would use the four grid points nearest the 2 degree target grid point. The nearby grid points could be all zero or include high values. In this case, conservative interpolation would be a good approach. Computing the curl of the wind stress requires that highly accurate derivatives be computed. In this case, patch interpolation would be appropriate. Deriving a new grid for categorical data (eg, land surface type) would be best accomplished with a nearest neighbor algorithm because interpolating different categories may result in a totally bogus result.

Miscellaneous Regridding Comments

Non-linear quantities should always be computed on the original grid and, subsequently, interpolated to the destination grid. Note that the results will be different than if the original variables on the source grid were interpolated to the destination grid and then the non-linear computations performed.

Vector interpolation (eg, U, V) should be performed on the vector pair simultaneously. Interpolating U, then separately, interpolating V may be adequate for some purposes, but not if the interpolated vector components were subsequently used to derive (say) divergence. An indirect approach would be to (a) calculate the scalar quantities vorticity

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and divergence on the source grid; (b) interpolate these scalar quantities to the destination grid using standard methods; and then, (c) derive the U and V from the rotational and divergent wind components.

Extrapolation: In some cases, a regridding algorithm may extrapolate rather than interpolate. A common example is extrapolating vertical profiles in mountain regions to (say) 1000hPa. In some cases, the extrapolation can be guided by physical principles. For example, using the standard lapse rate may be acceptable when extrapolating temperature or geopotential via the hydrostatic equation. However, generally, any extrapolated values should be used and interpreted with the utmost caution.

9. Taylor Diagrams

Taylor diagrams (Taylor, 2001) provide a way of graphically summarizing how closely a pattern (or a set of patterns) matches observations. The similarity between two patterns is quantified in terms of their correlation, their centered root-mean-square difference and the amplitude of their variations (represented by their standard deviations). These diagrams are especially useful in evaluating multiple aspects of complex models or in gauging the relative skill of many different models (e.g., IPCC, 2001).

Figure 1 is a sample Taylor diagram which shows how it can be used to

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summarize the relative skill with which several global climate models simulate the spatial pattern of annual mean precipitation. Statistics for eight models were computed, and a letter was assigned to each model considered. The position of each letter appearing on the plot quantifies how closely that model's simulated precipitation pattern matches observations. Consider model F, for example. Its

Figure 1: Sample Taylor diagram displaying a statistical comparison with

observations of eight model estimates of the global pattern of annual mean precipitation.

pattern correlation with observations is about 0.65. The centered root-mean-square (RMS) difference between the simulated and observed patterns is proportional to the distance to the point on the x-axis

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identified as "observed." The green contours indicate the RMS values and it can be seen that in the case of model F the centered RMS error is about 2.6 mm/day. The standard deviation of the simulated pattern is proportional to the radial distance from the origin. For model F the standard deviation of the simulated field (about 3.3 mm/day) is clearly greater than the observed standard deviation which is indicated by the dashed arc at the observed value of 2.9 mm/day.

The relative merits of various models can be inferred from figure 1. Simulated patterns that agree well with observations will lie nearest the point marked "observed" on the x-axis. These models will have relatively high correlation and low RMS errors. Models lying on the dashed arc will have the correct standard deviation (which indicates that the pattern variations are of the right amplitude). In figure 1 it can be seen that models A and C generally agree best with observations, each with about the same RMS error. Model A, however, has a slightly higher correlation with observations and has the same standard deviation as the observed, whereas model C has too little spatial variability (with a standard deviation of 2.3 mm/day compared to the observed value of 2.9 mm/day). Of the poorer performing models, model E has a low pattern correlation, while model D has variations that are much larger than observed, in both cases resulting in a relatively large (~3 mm/day) centered RMS error in the precipitation fields. Note also that although models D and B have about the same correlation with observations, model B simulates the amplitude of the

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variations (i.e., the standard deviation) much better than model D, and this results in a smaller RMS error.

In general, the Taylor diagram characterizes the statistical relationship between two fields, a "test" field (often representing a field simulated

by a model) and a "reference" field (usually representing "truth", based on observations). Note that the means of the fields are subtracted out before computing their second-order statistics, so the diagram does not provide information about overall biases, but solely characterizes the centered pattern error.

There are several minor variations on the diagram that have been found useful for various purposes (see, Taylor, 2001). For example,

- i. The diagram can be extended to a second "quadrant" (to the left) to allow for negative correlations.
 - ii. The statistics can be normalized (and non-dimensionalized), dividing both the RMS difference and the standard deviation of the "test" field by the standard deviation of the observations. In this case the "observed" point is plotted on the x-axis at unit distance from the origin. This makes it possible to plot statistics for different fields (with different units) on the same plot.
 - iii. The isolines drawn on the sample plot above are often omitted to make it easier to see the plotted points.
 - iv. When comparing fields simulated by two different versions of a
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model, the two points on the graph representing those fields are often connected by an arrow to indicate more clearly whether or not the model is moving toward "truth," as defined by observations.

10. Text (ASCII) files

Text files encountered in climate research are typically structured as a sequence of lines terminated with a special 'end-of-line' character (typically, the "null" character). Most climate oriented text files are in ASCII format. An advantage of text files is that they are human readable. Unfortunately, text files are very inefficient for archival or transmission of data. As a simple example, consider $x = -123.456$. In floating point binary form, this is stored as four bytes (32 bits) while in ASCII/text format, this takes eight bytes (64 bits) because each ASCII character requires eight bits.

11. Trend Analysis

The detection, estimation and prediction of trends and associated

statistical and physical significance are important aspects of climate research. Given a time series of (say) temperatures, the trend is the rate at which temperature changes over a time period. The trend may be linear or non-linear. However, generally, it is synonymous with the linear slope of the line fit to the time series. Simple linear regression is most commonly used to estimate the linear trend (slope) and statistical

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significance (via a Student-t test). The null hypothesis is no trend (ie, an unchanging climate). The non-parametric (ie., distribution free) Mann-Kendall (M-K) test can also be used to assess monotonic trend (linear or non-linear) significance. It is much less sensitive to outliers and skewed distributions. (Note: if the distribution of the deviations from the trend line is approximately normally distributed, the M-K will return essentially the same result as simple linear regression.) The M-K test is often combined with the Theil-Sen robust estimate of linear trend. Whatever test is used, the user should understand the underlying assumptions of both the technique used to generate the estimates of trend and the statistical methods used for testing. For example, the Student t-test assumes the residuals have zero mean and constant variance. Further, a time series of N values may have fewer than N independent values due to serial correlation or seasonal effects. The estimate of the number of independent values is sometimes called the equivalent sample size. There are methodologies to estimate the number of independent values. It is this value that should be used in assessing the statistical significance in the (say) Student t-test. Alternatively, the series may be pre-whitened or deseasonalized prior to applying the regression or M-K test statistical tests.

There are numerous caveats that should be kept in mind when analyzing trend. Some of these include: (1) Long term, observationally based estimates are subject to differing sampling networks. Coarser sampling is likely to result in larger uncertainties. Variables which have a large

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spatial autocorrelation (eg, temperature, sea level pressure) may have smaller sampling errors than (say) precipitation which generally has lower spatial correlation; (2) The climate system within which the

observations are made is not stationary; (3) Station, ship and satellite observations are subject to assorted errors. These could be random, systematic and external such as changing instruments, observation times or observational environments. Much work has been done on creating time series that takes into account these factors; (4) While reanalysis projects provide unchanging data assimilation and model frameworks, the observational mix changes over time. That may introduce discontinuities in the time series that may cause a trend to be estimated significant when in fact it is an artifact of the discontinuities; (5) Even a long series of random numbers may have segments with short term trends. For example, the well known surface temperature record from the Climate Research Unit which spans 1850-present, shows an undeniable long-term warming trend. However, there are short term negative trends of 10-15 years embedded within this series. Also, the rate of warming changes depending on the starting date used in that time series; (6) As noted above, a series on N observations does not necessarily mean these observations are independent. Often, there is some temporal correlation. This should be taken into account for example when computing the degrees of freedom of the t-test.

Web Based Visualization and Processing for Climate and Weather

There are number of web-based sites that provide the cabability to visualize, perform limited proceessing tasks and, somtimes download data. Some are

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designed for a specific data product while others are cabable of creating graphics and performing computations on multible data sets. Common tasks include difference maps, correlations, composites, climate monitoring and deriving climate indices. Often generated graphical images can be downloaded using a variety of file image formats. Downsides of these tools include little, if any, ability to alter the preset graphical images or expand the available computational capabilities or add additional data sets.

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Paper 6 of 7 — EMT 226

Level 200 • Second Semester • Handout

Introductory Climatology and Biogeography (EMT 226)

Introductory Climatology

Definitions

Climatology is the science of climate which studies the Physical state of the atmosphere over a specific region during a specific period on the basis of climatic data. Climatology is compounded of two Greek words, Klima and Logos. Klima meaning inclination, that is latitude, Logos meaning science of study. So, climatology is a science that seeks to describe and explain the nature of climate, why it differs from place to place and how it is related to other elements of the natural environment and human activities. It is the study of the verities of climates found on the Earth and their distribution over the surface of the Earth.

Climatology is that science which studies the nature of climate, the causes and interpretation of its spatial variations and its association with elements of natural environment and human activities.

Climatology is the study of climate, its variations and extremes, and its influences on a variety of activities including (but far from limited to) human health, safety and welfare. Climate, in a narrow sense, can be defined as the average weather conditions for a particular location over a period of time.

Climate can be described in terms of statistical descriptions of the central tendencies and variability of relevant elements such as temperature, precipitation, atmospheric pressure, humidity, and winds, or through combinations of elements, such as weather types and phenomena, that are typical to a location, region or the world for a given time period.

Climatology is the study of Earth's weather over long enough periods of time to allow for averages and trends to be developed. It investigates atmospheric changes (both long and short-term) that define average climates and their change over time, due to both natural and anthropogenic climate variability.

We can therefore conclude that climatology is the scientific study of climate, specifically of weather condition trends over a defined period of time, past, present or future. In analyzing long-term climatic developments, climatology is distinct from meteorology, which is associated with short-term weather system studies.

Nature and Scope of Climatology

Climate is defined as weather patterns that have been averaged over a given period of time, at least 30 years, to obtain a consistent pattern of the expected atmospheric conditions. Weather is the atmospheric condition of a particular place over a short period of time, normally a day.

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Weather averaging for a long and indefinite period of time makes it possible to predict the

climatic pattern of an area.

Climatology is a branch of the atmospheric sciences and a subfield of physical geography, which is one of the Earth Sciences. Climatology includes aspects of oceanography, hydrology and biogeochemistry. It focuses on aspects such as atmospheric boundary layer, circulation patterns, heat transfer in the globe, ocean interaction with the atmosphere and land surface, land use and topography. Basic knowledge of climate can be used within shorter term weather forecasting while climate models are used for a variety of purposes from study of the dynamics of the weather and climate system to projections of future climate.

Climatology has evolved from a simple theoretical bookkeeping activity to the current complex multidisciplinary scientific and a practical field. Climatology is not only concerned with the climate of a place but it also establishes the reason for the fluctuation of climate in the area, how human activities lead to climatic variations, effects of the climate on human activities, and the characteristics of the climate.

Branches of Climatology

Climatology is broken into several major sub-fields. Categorization of climatology depends on various criteria. Common fields include, but are not limited to:

Historical Climatology – The study of climate in relationship to human history using evidence like direct temperature recording, agricultural practices, or indirect historical evidence.

Paleoclimatology – Reconstruction of past climates using fossil evidence, ice cores, and tree rings.

Boundary-layer climatology is primarily concerned with exchanges in energy, mass, and momentum near the surface.

Hydroclimatology involves the processes (at all scales) of interaction between the atmosphere and near-surface water in all of its forms.

Bioclimatology is a very diverse sub-discipline that includes the interaction of living things with their atmospheric environment.

Physical Climatology deals with the interpretation of factors responsible for the spatial and temporal variations of exchange of air circulations, heat and humidity (moisture). It studies various elements of weather namely insolation, temperature, precipitation, fogs, visibility etc. and complicated patterns and exchange of energy from one phase to another- from solid state to liquid state and from liquid state to vapor and vice versa.

Dynamic climatology is global in scope. It studies the thermodynamic processes in the atmosphere and the resultant atmospheric motions. It investigates the impact of changes in

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various physical parameters on climate. This branch of climatology includes (i) the effect of the increase of greenhouse gases in global temperature (ii) the role of mountains in determining the dynamics of the atmosphere (iii) an investigation into sea-surface temperature.

Synoptic climatology deals with local or hemispheric climate from the view point of atmospheric circulation (different circulation patterns lead to differences in climates). It studies the

relationship between circulation features and severe weather conditions e.g. the effect of El Nino and La Nina in creating severe weather conditions

Regional climatology seeks to determine and describe the various types of world climates. It is also known as descriptive climatology because it is concerned with the identification of important climatic characteristics and the interaction of weather and climatic elements upon the life, health and economic conditions of the people and areas. Regional climatology includes the study of types, distribution, pattern, characteristics and factors of world climates. Regional climatology is studied in three ways on the basis of spatial scale viz:

1. Macro-climatic region
2. Meso-climatic region
3. Micro-climatic region

Applied climatology studies the application of climatic knowledge to solve various problems faced by human society e.g. global warming and climate changes. It studies the interactions between climate and biosphere i.e. how does climate influence and control plants and animals including man and in turn how does man modify climate by introducing advertent and inadvertent changes in the physical environment and by making certain weather modification e.g. cloud seeding and induced precipitation. Further, applied climatology includes the study of variability of climate, climatic changes, air pollution, climate and comfort, climate and health, climate and society, extreme weather events (e.g. floods, drought etc.) and their impacts, climate and recreation, climate and urban planning, weather forecasting, climate and transport and communication, weather modification etc.

Interrelationship between Meteorology and Climatology

Meteorology studies changes in weather, the state of atmospheric properties for a given location, while climatology examines weather properties over time for a location.

Meteorologists focus on forecasting weather. They look at a few basic atmospheric interactions and hope to predict the weather in the next few days. Climatologists take a much larger view of the whole climate/weather idea. They look at how climates are created and what they do to the environment. Climatology is a long-term study of the geographic world.

Certainly, climatology is closely related to meteorology. Meteorology deals with the day to day atmospheric conditions and their causes and sometimes referred to as the Physics of the atmosphere. Meteorology uses the methods of Physical science to interpret and explain the

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atmospheric processes. Climatology on other hand collects and interprets the data observed by meteorology to investigate the spatial patterns of climate and its interaction. Climatology studies the atmospheric conditions over time and the spatial distribution of these climatic conditions. As such, Climatology is a science whose methods are strictly meteorological and whose aims and results are geographical as it draws its knowledge from Meteorology and sifts the weather data to make geographical statements.

What this means is that rather than study the immediate weather and what its impact is or will be in the next day or two as meteorologists do, climatologists study long-term trends in the Earth's

weather and try to understand what the impact of various things, including human activity, will be over decades or centuries.

Both climatology and meteorology are branches of physical science that deal with the weather. While they are related to one another in many ways, they aren't the same thing. As the saying goes, the weather changes frequently, but the climate is relatively stable. That saying defines the major difference between the two sciences, which is basically one of time.

History of Climatology

Early references to the weather can be found in the poems of ancient Greece and in the Old Testament of the Judaeo-

Christian Bible. Even older references can be singled out in the Vedas, the most ancient Hindu scriptures that were written about 1800 BCE. Specific writings on the theme of meteorology and climatology are found in Hippocrates' Air, Waters, and Places dated around 400 BCE followed by Aristotle's Meteorologica written around 350 BCE. To the early Greek philosophers, climate meant "slope" and referred to the curvature of the Earth's surface which, due to the changing incidence of the sun's rays, gives rise to the variation of climate with latitude. Logical and reliable inferences on climate can be found in the works of the Alexandrian philosophers Eratosthenes and Aristarchus.

With the onset of extensive geographical exploration in the 15th century, descriptions of the earth's climates and the conditions giving rise to the climates started to emerge. The invention of meteorological instruments such as the thermometer in 1593 by Galileo Galilei and the barometer in 1643 by Evangelista Torricelli gave a greater impulse to the establishment of mathematical and physical relationships between the different characteristics of the atmosphere and, therefore, of relationships which could describe the state of the climate at different times and in different places.

The observed pattern of circulation linking the tropics and subtropics, including the trade winds, tropical convection, and subtropical deserts, was first interpreted by George Hadley in 1735, and thereafter known as the Hadley Cell. Julius von Hann, who published the first of three volumes of the Handbook of Climatology in 1883, wrote the classic work

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on general and regional climatology that included data and eyewitness descriptions of weather and climate.

In 1918 Wladimir Köppen produced the first detailed classification of world climates, based on the vegetative cover of land. This endeavor was followed by more detailed development in descriptive climatology. The geographer E.E. Federov, for example, attempted to describe local climates in terms of daily weather observations.

In the first thirty years of the 20th Century the diligent and combined use of global observations and mathematical theory to describe the atmosphere, led to the identification of large scale atmospheric patterns. Notable in this field was Sir Gilbert Walker, who conducted detailed studies of the Indian monsoon, the Southern Oscillation, the North Atlantic Oscillation, and the North Pacific Oscillation.

Other major works on climatology included those by Tor Bergeron (on dynamic climatology in 1928) and Wladimir Köppen and Rudolf Geiger (a climatology handbook in 1936). Geiger first described the concept of microclimatology in some detail in 1927, but the development of this field did not occur until the Second World War. During the war, for planning purposes, a probability risk concept of weather data for months or even years ahead was found to be necessary and was tried. C.W. Thornthwaite established a climate classification in 1948 based on a water budget and evapotranspiration. In the following decades development of theories on climatology saw major progress.

During the latter decades of the 20th century the issue of climate change began to focus attention on the need to understand climate as a major part of a global system of interacting processes involving all of the earth's major domains. Climate change is defined as a statistically significant variation in either the average state of the climate or in its variability, persisting for an extended period, typically decades or longer. It may be caused by either natural internal processes, external forcing, or by persistent anthropogenic (resulting from or produced by human activity) changes in the composition of the atmosphere or in land use.

Climatology Investigation Methods

There are different approaches in Climatological investigations.

- Paleoclimatology which seeks to reconstruct past climates through examining records such as ice cores and tree rings (dendroclimatology);
- Use of meteorological data accumulated over many years to study contemporary climates;
- Use of statistical or mathematical models to study the atmosphere and its dynamics;
- Modeling for understanding past, present and potential future climates; and

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- Historical climatology where climates are related to human history and thus focuses only on the last few thousand years.

Properties

Three properties of the climate include "normals," "extremes," and "frequencies" and are used to gauge the state of the atmosphere over a particular time period

- "Normals" refers to average weather conditions at a place
- "extremes" are used to describe the maximum and minimum measurements of atmospheric variables
- "frequencies" refers to the rate of incidence of a particular phenomenon at a particular place, over a long period of time

Since climate is a long term representation of weather on a large scale and is a product of complex processes, researchers need to have very good understanding of physical laws usually expressed as differential equations. This makes it difficult to conduct climate research since the

equations are normally coupled and are non-linear, thus requiring numerical methods to obtain approximate solutions. Sometimes climate is modeled as a stochastic process when approximating very complex processes. These activities necessitate the quantitative approach in climatology.

The Climate System

The climate system is a complex, interactive system consisting of the atmosphere, land surface, snow and ice, oceans and other bodies of water, and living organisms. The atmosphere is the gaseous envelope surrounding the Earth. The dry atmosphere consists almost entirely of nitrogen and oxygen, but also contains small quantities of argon, helium, carbon dioxide, ozone, methane, and many other trace gases. The atmosphere also contains water vapor, condensed water droplets in the form of clouds, and aerosols.

The hydrosphere is that part of the Earth's climate system comprising liquid water distributed on and beneath the Earth's surface in oceans, seas, rivers, freshwater lakes, underground reservoirs, and other water bodies. The cryosphere collectively describes elements of the Earth system containing water in its frozen state and includes all snow and ice (sea ice, lake and river ice, snow cover, solid precipitation, glaciers, ice caps, ice sheets, permafrost, and seasonally frozen ground). The surface lithosphere is the upper layer of the solid Earth, both continental and oceanic. The biosphere comprises all ecosystems and living organisms in the atmosphere, on land (terrestrial biosphere), or in the oceans (marine biosphere), including derived dead organic matter, such as litter, soil organic matter, and oceanic detritus.

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The climate system

Under the effects of solar radiation and the radiative properties of the surface, the climate of the Earth is determined by interactions among the components of the climate system. The interaction of the atmosphere with the other components plays a dominant role in forming the climate. The atmosphere obtains energy directly from incident solar radiation or indirectly via processes involving the Earth's surface. This energy is continuously redistributed vertically and horizontally through thermodynamic processes or large scale motions with the unattainable aim of achieving a stable and balanced state of the system. Water vapor plays a significant role in the vertical redistribution of heat by condensation and latent heat transport. The ocean, with its vast heat capacity, limits the rate of temperature change in the atmosphere and supplies water vapor and sensible heat to the atmosphere. The distribution of the continents affects oceanic currents, and mountains redirect atmospheric motions. The polar, mountain, and sea ice reflect solar radiation back to space. In high latitudes the sea ice acts as an insulator and protects the ocean from rapid energy loss to the much colder atmosphere. The biosphere, including its human activities, affects atmospheric components such as carbon dioxide as well as features of the Earth's

surface such as soil moisture and albedo.

Interactions among the components occur on all scales Spatially, the microscale encompasses features of climate characteristics over small areas such as individual buildings and plants or fields. A change in microclimate can be of major

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importance when the physical characteristics of an area change. New buildings may produce extra windiness, reduced ventilation, excessive runoff of rainwater, and increased pollution and heat. Natural variations in microclimate, such as those related to shelter and exposure, sunshine and shade, are also important, for example, from determining which plants will prosper in a particular location, to the need to make provision for safe operational work and leisure activities.

The mesoscale encompasses the climate of a region of limited extent, such as a river catchment area, valley, conurbation, or forest. Mesoscale variations are important in applications including land use, irrigation and damming, the location of natural energy facilities, and resort location.

The macroscale encompasses the climate of large geographical areas, continents, and the globe. It determines national resources and constraints in agricultural production and water management, and is thus linked to the nature and scope of human health and welfare. It also defines and determines the impact of major features of the global circulation such as the El Niño Southern Oscillation (ENSO), the monsoons, and the North Atlantic Oscillation.

A temporal scale is an interval of time. It can range from minutes and hours to decades, centuries, and longer. The characteristics of an element over an hour are important, for example, in agricultural operations such as pesticide control and in monitoring energy usage for heating and cooling. The characteristics of an element over a day might determine, for example, the human activities that be safely pursued. The climate over months or years will determine, for example, the crops that can be grown or the availability of drinking water and food. Longer time scales of decades and centuries are important for studies of climate variation caused by natural phenomena such as atmospheric and oceanic circulation changes and by the activities of humans.

Contributions of Climatology

Climatology has become a dynamic and highly functional branch of science. New techniques are being developed and investigations undertaken to study the application of climate in many fields such as agriculture, forestry, ecosystems, energy, industry, production and distribution of goods, engineering design and construction, human well-being, transportation, tourism, insurance, water resource and disaster management, fisheries, and coastal development.

The crucial roles of climate data and climate predictions in planning for disaster mitigation and sustainable development, and in addressing all the consequences of climate change are now firmly established within various conventions such as the United Nations Framework Convention on Climate Change.

Applied climatology makes the maximum use of meteorological and climatological

knowledge and information for solving practical social, economic, and environmental problems. Climatological services are designed for a variety of public, commercial, and industrial users. Assessments of the effects of climate variability and climate change on

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human activities, as well as the effects of human activities on climate, are major factors in local, national and global economic development, social programmes, and resource management.

Current interest on the consequences of economic and human activities on climate and how climate variability and change influence human societies highlights the need for further research into the physical and dynamical processes in the climate system and for their statistical description. Understanding of natural climate variability, appreciation of climate sensitivity to human activities, and insight on the predictability of the weather and climate for periods ranging from days to decades is fundamental to improving our capability to respond effectively to economic and societal problems.

Precisely, fundamental importance of climatology can be highlighted as follows:

- Climatology seeks to explain the causes of different types of climates
 - o The reasons for their variations
 - o Their effects on natural vegetation
 - o The processes that produce different climates
- Climatology makes a detail analysis of the interaction of weather and climatic elements upon human societies
- Climatology discusses the various climatic elements, the factors that control the distribution of climate over the Earth
- Climatology is important in determining the climatic patterns of a particular region. Establishing the climatic pattern is significant in deciding the economic activities that would thrive in that particular region. If the climate of a region is established to be cool and wet, it would be safe to conclude that agriculture might thrive in the region.
- Having a clear climate pattern makes it easier for people to understand the seasons of engaging in particular tasks. This is especially most important to tourists and farmers.
- Infrastructure development, especially buildings are dependent on climate. After a climatic pattern has been established, the engineers recommend the use of materials that would not only withstand the conditions but also protect the dwellers from any harsh climatic condition.

Contemporary Climatological Challenges

Climate change has become a major issue for the human community. Human activities, especially the burning of fossil fuels, have led to changes in the composition of the global atmosphere. Marked increases in tropospheric carbon dioxide and methane during the industrial era, along with increased aerosols and particulate emissions are significantly affecting global climate. Chlorofluorocarbons extensively used in the past as aerosol propellants, cleaning fluids, and refrigerants are a main cause of stratospheric ozone

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depletion. Over one fifth of the world's tropical forests were cleared between 1960 and 2000, likely altering the complex mesoscale and global hydrological cycles. Artificial canyons formed in cities by buildings, together with the asphalt of roads, increase the amount of radiation absorbed from the sun and form urban heat islands. Accelerated runoff of rainwater and the removal of trees and other vegetation reduce the amount of transpired water vapor that would otherwise help to moderate temperature. Pollution from vehicles and buildings accumulates, especially in calm conditions, and causes many human health problems and damage to structures.

Introductory Biogeography

Definitions

The term Biogeography is derived from the union of two words "Bio" and "Geography". Bio refers to biology, a science that deals with living phenomena while geography is a science that is concerned with distribution and interrelationships between both living and nonliving phenomena. Biogeography as a branch of physical geography is the study of spatial distribution of both plants and animals and the processes that produce variations in the patterns of distribution (Browne, 1993). Biogeography is considered as an interdisciplinary branch of geography and biology which deals with the spatial and temporal patterns of flora and fauna in respect of surviving geographical conditions.

Biogeography refers to the study of distribution of species and ecosystems in geographic space through geological time. Biogeography is often studied in the context of ecological and historical factors which have shaped the geographical distribution of organisms over time. Specifically, species vary geographically based on latitude, habitat, segregation/isolation (e.g., islands), and elevation.

Biogeography implies a linkage between Biology and Geography. It studies the distribution of biological materials over the earth's surface and the factors responsible for the observed spatial variations (Pears, 1977). Jones (1980) described Biogeography as a subject that studies the expansion pattern of organisms in the past and present.

Biogeography is the study of past and present-day spatial distributions of life (both flora and fauna) on earth in relation to environmental factors such as geology, climate and soils.

Nature and Scope

Biogeography enables us to identify biodiversity patterns in the past and present, identify the expansion of organisms which enables us to study about the relationship between living and non-living factors that influence organisms to exist. Diversity of flora and fauna exists according to the relationship that emerges with these factors.

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The field of biogeography covers the trio-spheres of the earth viz- hydrosphere, lithosphere and atmosphere. Combination of these three gives rise to a fourth sphere known as biosphere. More precisely it is the study of the biologically inhabited part of the Earth therefore known as

Biosphere (De Quiroz, 2005). Its subject matter covers the multitudinous forms of plant and animal life which inhabit „biosphere“ a relatively shallow but densely populated zone as well as the complex biological processes which control their activities.

Biogeography is an interdisciplinary natural science with a complex, even cryptic identity that defies definition. In essence, it is the science of the distribution of living organism and the factors that underlie this distribution. Not only is biogeography concerned with spatial patterns in the contemporary sense, but also there is a strong evolutionary component and a concern with change over time (Meadows, 2015).

Within biogeography there are various degrees of specialization, usually involving taxonomic study of either fauna (zoogeography) or flora (phytogeography). As such, biogeography depends heavily upon the observations and concepts of its parent subjects– geography, earth science and biology, and especially ecology and paleoecology (Pitman, 2011).

Summarily, nature and scope of biogeography can be outlined as:

- Biogeography studies the cooperative relationship between vegetation without intelligence and the man with intellectual capacity.
- Biogeography mainly pays its attention towards the diversities according to spatial changes.
- Biogeography studies different changes of animal life
- Biogeography studies how different types of animals get acclimatized to the environment. Changes in flora and fauna life take place according to the climate. As for animals, animals living in warm regions are different from animals living in cold regions.
- Biogeography studies the lithosphere; living beings of interior earth, there living begins, soil and rocks. It studies more on soil than rocks and also about mass movement of micro ecological systems.
- Attention is paid towards human effects in addition to physical and biological factors. Especially, the natural flora changes in agricultural process. (Eg: clearing of forests for cultivation, changes the ecological condition which existed before). Accordingly, biogeography studies about the human effects towards the environment.

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Branches

There are several branches/fields of biogeography and each addresses the distribution of species from a different perspective.

1. Phytogeography is the branch of biogeography that studies the distribution of plants.
2. Zoogeography is the branch that studies distribution of animals.
3. Historical biogeography primarily involves specie distributions from an evolutionary perspective. Studies of historical biogeography involve the investigation of phylogenic distributions over time.
4. Ecological biogeography refers to the study of the contributing factors for the global distribution of plant and animal species. Some examples of ecological factors that are commonly studied include climate, habitat and primary productivity (the rate at which the

plants in a particular ecosystem produce the net chemical energy). Moreover, ecological biogeography differs from historical biogeography in that it involves the short-term distribution of various organisms, rather than the long-term changes over evolutionary periods.

5. Conservation biogeography seeks to effectively manage the current level of biodiversity throughout the world by providing policymakers with data and potential concerns regarding conservation biology.

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Investigation Approaches

The major approaches in biogeography are classified broadly into those rooted in the spatial tradition and those associated with a more ecological theme, as shown in the schematic representation below (Meadows, 2015).

Spatial Tradition: Focus is on Phytogeography, Zoogeography, Historical Biogeography and Vicariance dispersal. Phytogeography and zoogeography are concerned with the geographic distribution of the plant species and animal species respectively. Historical biogeography study about how species distribution have changed overtime in relationship to the history of landforms, ocean basis and climate as well as how those changes have contributed to the evolution of biota. Vicariance dispersal means how species have been distributed and the division of a widespread group of organisms by a geographic barrier, such as a mountain range or a body of water, often resulting in the evolution of related species on either side of the barrier.

Ecological Tradition: Focus is on study of Ecosystem, Paleoecology and Island Biogeography. Ecology is the study of the interaction among organism and an ecosystem is comprised of habitats, biological communities, and ecotones (a transition area between two biomes or where two communities meet). Therefore to understand the distribution of species on earth a fundamental knowledge of ecology and ecosystems is essential. Paleoecology study fossil of animals and plants in order to realize their ecology and the environmental conditions in which

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they lived. Island biogeography seek to understand the generation of biodiversity on island archipelagos in relation to geographical isolation and dispersal abilities of taxa.

Comparative Biogeography: The study of comparative biogeography can follow two main lines of investigation; Systematic biogeography, the study of biotic area relationships, their distribution, and hierarchical classification and Evolutionary biogeography, the proposal of evolutionary mechanisms responsible for organismal distributions. Possible mechanisms include widespread taxa disrupted by continental break-up or individual episodes of long-distance movement.

In spite of the fact that this field of study is a transition between biology and geography; it is not the exclusive preserve of either of these two sciences. By its very character biogeography is situated at and overlaps the boundaries of a great number of other disciplines. Disciplines like geology, climatology, pedology, geomorphology as well as the botany, zoology, genetics and geography contribute to this very large and varied field. In doing so they are to a greater extent,

dependent on an understanding of biogeography. As a result, the approach to biogeography in large measure is determined by the interest and objectives of the particular scholar of a specific scientific field (Lomolino, 2000).

Thrust of Inquiry

The basic material for biogeography investigation consists of the organisms, plant and animal communities which exhibit wide extension over the globe with varying degrees of the biosphere which envelops the surface of the earth and consists of all living-organisms living on land, aquatic and aerial areas.

The patterns of species distribution across geographical areas can usually be explained through a combination of historical factors such as: speciation, extinction, continental drift, and glaciation. Through observing the geographic distribution of species, we can see associated variations in sea level, river routes, habitat, and river capture. Additionally, this science considers the geographic constraints of landmass areas and isolation, as well as the available ecosystem energy supplies. Biogeography explores questions on whether the situation of identifying the life styles that existed in the past are being practiced in the present, whether those practices are being followed in past have been subjected to changes and what the factors that influenced such changes. It is important to point out that in both biological and geography literature attention has been given to primarily though not exclusively concerned with the study of plant rather than animal geography. There are various reasons for this. Greater mobility combined with the small size and an elusive habit of life of the majority of animal species make a study of their distribution more difficult than that of plants. Also, until fairly recent, zoogeography had not developed to anything like the same extent as plant geography. Other considerations have however, been responsible for the geographer"s preoccupation with plant life. Plants-vegetable matter-both living and decaying, comprise the greatest bulk of the total world biomass both above and below

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the ground surface and in water bodies. Hence plants are the most conspicuous components of the biosphere and en masse major landscape element. Further, plants are more directly depend upon their physical habitat than animals. In comparison to the latter, plants are relatively immobile or stable due to lack of locomotive organs. As a result they provide a better index of the total environment -physical, biological and human. Because of their greater biomass, plants exert a greater influence on the character of the atmosphere and soil they occupy. They not only modify the physical habitat, but in doing so they create a particular biological environment which would not otherwise exist. Finally plants provide the primary source of food -energy for all other living organisms-including man. They are basic to the geography of animals and constitute the most important of man"s resources (Holden et. al., 2004).

Also, biogeography is most keenly observed on the world's islands. These habitats are often much more manageable areas of study because they are more condensed than larger ecosystems on the mainland. Islands are also ideal locations because they allow scientists to look at habitats that new invasive species have only recently colonized and can observe how they disperse throughout the island and change it. They can then apply their understanding to similar but more complex mainland habitats. Islands are very diverse in their biomes, ranging from the tropical to

arctic climates. This diversity in habitat allows for a wide range of species study in different parts of the world.

Modern biogeography often employs the use of Geographic Information Systems (GIS), to understand the factors affecting organism distribution, and to predict future trends in organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

As such, in terms of applications of biogeography as a science today, technological advances have allowed satellite imaging and processing of the Earth. Two main types of satellite imaging that are important within modern biogeography are Global Production Efficiency Model (GLO-PEM) and Geographic Information Systems (GIS). GLO-PEM uses satellite-imaging gives "repetitive, spatially contiguous, and time specific observations of vegetation". These observations are on a global scale. GIS show certain processes on the earth's surface e.g. whale locations, sea surface temperatures, and bathymetry (Underwater – ocean, sea, lakes, rivers etc. - depths and topography) etc. Current scientists also use coral reefs to delve into the history of biogeography through the fossilized reefs.

Basic Concepts

Some fundamental concepts in biogeography include:

□ **Species** – A species is a group of organisms that share a genetic heritage, are able to interbreed, and to create offspring that are also fertile. A species is often defined as the largest group of organisms in which two individuals can produce fertile offspring, typically by sexual reproduction. A grouping of the same type of organisms.

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□ **Populations** – A group of species in a specific area at a specific time. Population is all the organisms of the same group or species, which live in a particular geographical area, and have the capability of interbreeding. A group of individuals of the same species occupying a particular geographic area.

□ **Communities** – All populations in a specific area at a specific time. In ecology, a community is a group or association of populations of two or more different species occupying the same geographical area and in a particular time.

□ **Ecosystems** – All living and non-living things in their environment as well as their interactions. An ecosystem includes all of the living things (plants, animals and organisms) in a given area, interacting with each other, and also with their non-living environments (weather, earth, sun, soil, climate, atmosphere) - food chain, food web, nutrient cycles

□ **Ecozones** - An ecozone (or biogeographic realm) is the broadest biogeographic division of the Earth's land (or water) surface, based on distributional patterns of terrestrial (or aquatic) organisms. They are subdivided in ecoregions, which are classified in biomes or habitat types. Each ecozone may include a number of different biomes. E.g. terrestrial ecozones include Nearctic, Afrotropics and Oceania realms, marine include eastern pacific, western Atlantic regions etc.

□ **Ecoregions** – Large unit of land or water containing a geographically distinct assemblage

of species, natural communities and environmental conditions. They are a division of the larger "ecozones" or biogeographic regions. E.g. desert, forest (mangrove, rainforest), grassland (guinea, sudan, sahel) etc.

□ Ecological succession – The non-seasonal change of plant communities over time.

Primary (e.g. volcanic explosion) & Secondary (hurricane). Pioneer species (first plant species to return to an area during primary succession)

□ Allopatric Speciation – This is the splitting of a species by evolution of geographically isolated populations

□ Evolution – change in genetic composition of a population

□ Extinction – disappearance of a species

□ Dispersal – movement of populations away from their point of origin, related to migration

□ Endemic Areas – Locations or populations where a specie exists permanently

□ Geodispersal – The erosion of barriers to biotic dispersal and gene flow, that permit range expansion and the merging of previously isolated biotas

□ Range and Distribution - The range of a species is the geographical area within which that species can be found. Within that range, distribution is the general structure of the species population

□ Vicariance –It is the separation or division of a group of organisms by a geographic barrier, such as a mountain or a body of water, resulting in differentiation of the original group into new varieties or species. This can also be referred to as fragmentation of the

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environment or geographical separation typically by a physical barrier (such as a mountain range, river or splitting of a tectonic plate) in contrast to dispersal as a factor in promoting biological evolution by division of large populations into isolated subpopulations.

Link between Biogeography and Climatology

Climatology deals with study of various elements and interrelated weather phenomena which affects life on the earth. Under normal circumstances climate is essential to the growth of plants and animals including human beings but in adverse circumstances like drought and flood it disturbs the lives and their habitats. Therefore climate is an inseparable part of biogeography. Its influence can be seen in some biogeographical themes such as Agriculture, Physiological adaptation, Dispersal, Range etc.

Applied climatology studies the application of climatic knowledge to solve various problems faced by human society e.g. global warming and climate changes. It studies the interactions between climate and biosphere i.e. how does climate influence and control plants and animals including man and in turn how does man modify climate by introducing advertent and inadvertent changes in the physical environment and by making certain weather modification e.g. cloud seeding to induce precipitation.

Generally, atmospheric processes particularly the climate significantly influences the biotic and

abiotic components of the terrestrial and aquatic ecosystems. Therefore, problem solving in environmental issues has to adopt both interdisciplinary and multidisciplinary approach.

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Paper 7 of 7 — EMT 226

Level 200 • Second Semester • Handout

Introductory Climatology and Biogeography (EMT 226)

Introductory Climatology

Definitions

Climatology is the science of climate which studies the Physical state of the atmosphere over a specific region during a specific period on the basis of climatic data. Climatology is compounded of two Greek words, Klima and Logos. Klima meaning inclination, that is latitude, Logos meaning science of study. So, climatology is a science that seeks to describe and explain the nature of climate, why it differs from place to place and how it is related to other elements of the natural environment and human activities. It is the study of the verities of climates found on the Earth and their distribution over the surface of the Earth.

Climatology is that science which studies the nature of climate, the causes and interpretation of its spatial variations and its association with elements of natural environment and human activities.

Climatology is the study of climate, its variations and extremes, and its influences on a variety of activities including (but far from limited to) human health, safety and welfare. Climate, in a narrow sense, can be defined as the average weather conditions for a particular location over a period of time.

Climate can be described in terms of statistical descriptions of the central tendencies and variability of relevant elements such as temperature, precipitation, atmospheric pressure, humidity, and winds, or through combinations of elements, such as weather types and phenomena, that are typical to a location, region or the world for a given time period.

Climatology is the study of Earth's weather over long enough periods of time to allow for averages and trends to be developed. It investigates atmospheric changes (both long and short-term) that define average climates and their change over time, due to both natural and anthropogenic climate variability.

We can therefore conclude that climatology is the scientific study of climate, specifically of weather condition trends over a defined period of time, past, present or future. In analyzing long-term climatic developments, climatology is distinct from meteorology, which is associated with short-term weather system studies.

Nature and Scope of Climatology

Climate is defined as weather patterns that have been averaged over a given period of time, at least 30 years, to obtain a consistent pattern of the expected atmospheric conditions. Weather is the atmospheric condition of a particular place over a short period of time, normally a day.

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Weather averaging for a long and indefinite period of time makes it possible to predict the

climatic pattern of an area.

Climatology is a branch of the atmospheric sciences and a subfield of physical geography, which is one of the Earth Sciences. Climatology includes aspects of oceanography, hydrology and biogeochemistry. It focuses on aspects such as atmospheric boundary layer, circulation patterns, heat transfer in the globe, ocean interaction with the atmosphere and land surface, land use and topography. Basic knowledge of climate can be used within shorter term weather forecasting while climate models are used for a variety of purposes from study of the dynamics of the weather and climate system to projections of future climate.

Climatology has evolved from a simple theoretical bookkeeping activity to the current complex multidisciplinary scientific and a practical field. Climatology is not only concerned with the climate of a place but it also establishes the reason for the fluctuation of climate in the area, how human activities lead to climatic variations, effects of the climate on human activities, and the characteristics of the climate.

Branches of Climatology

Climatology is broken into several major sub-fields. Categorization of climatology depends on various criteria. Common fields include, but are not limited to:

Historical Climatology – The study of climate in relationship to human history using evidence like direct temperature recording, agricultural practices, or indirect historical evidence.

Paleoclimatology – Reconstruction of past climates using fossil evidence, ice cores, and tree rings.

Boundary-layer climatology is primarily concerned with exchanges in energy, mass, and momentum near the surface.

Hydroclimatology involves the processes (at all scales) of interaction between the atmosphere and near-surface water in all of its forms.

Bioclimatology is a very diverse sub-discipline that includes the interaction of living things with their atmospheric environment.

Physical Climatology deals with the interpretation of factors responsible for the spatial and temporal variations of exchange of air circulations, heat and humidity (moisture). It studies various elements of weather namely insolation, temperature, precipitation, fogs, visibility etc. and complicated patterns and exchange of energy from one phase to another- from solid state to liquid state and from liquid state to vapor and vice versa.

Dynamic climatology is global in scope. It studies the thermodynamic processes in the atmosphere and the resultant atmospheric motions. It investigates the impact of changes in

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various physical parameters on climate. This branch of climatology includes (i) the effect of the increase of greenhouse gases in global temperature (ii) the role of mountains in determining the dynamics of the atmosphere (iii) an investigation into sea-surface temperature.

Synoptic climatology deals with local or hemispheric climate from the view point of atmospheric circulation (different circulation patterns lead to differences in climates). It studies the

relationship between circulation features and severe weather conditions e.g. the effect of El Nino and La Nina in creating severe weather conditions

Regional climatology seeks to determine and describe the various types of world climates. It is also known as descriptive climatology because it is concerned with the identification of important climatic characteristics and the interaction of weather and climatic elements upon the life, health and economic conditions of the people and areas. Regional climatology includes the study of types, distribution, pattern, characteristics and factors of world climates. Regional climatology is studied in three ways on the basis of spatial scale viz:

1. Macro-climatic region
2. Meso-climatic region
3. Micro-climatic region

Applied climatology studies the application of climatic knowledge to solve various problems faced by human society e.g. global warming and climate changes. It studies the interactions between climate and biosphere i.e. how does climate influence and control plants and animals including man and in turn how does man modify climate by introducing advertent and inadvertent changes in the physical environment and by making certain weather modification e.g. cloud seeding and induced precipitation. Further, applied climatology includes the study of variability of climate, climatic changes, air pollution, climate and comfort, climate and health, climate and society, extreme weather events (e.g. floods, drought etc.) and their impacts, climate and recreation, climate and urban planning, weather forecasting, climate and transport and communication, weather modification etc.

Interrelationship between Meteorology and Climatology

Meteorology studies changes in weather, the state of atmospheric properties for a given location, while climatology examines weather properties over time for a location.

Meteorologists focus on forecasting weather. They look at a few basic atmospheric interactions and hope to predict the weather in the next few days. Climatologists take a much larger view of the whole climate/weather idea. They look at how climates are created and what they do to the environment. Climatology is a long-term study of the geographic world.

Certainly, climatology is closely related to meteorology. Meteorology deals with the day to day atmospheric conditions and their causes and sometimes referred to as the Physics of the atmosphere. Meteorology uses the methods of Physical science to interpret and explain the

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atmospheric processes. Climatology on other hand collects and interprets the data observed by meteorology to investigate the spatial patterns of climate and its interaction. Climatology studies the atmospheric conditions over time and the spatial distribution of these climatic conditions. As such, Climatology is a science whose methods are strictly meteorological and whose aims and results are geographical as it draws its knowledge from Meteorology and sifts the weather data to make geographical statements.

What this means is that rather than study the immediate weather and what its impact is or will be in the next day or two as meteorologists do, climatologists study long-term trends in the Earth's

weather and try to understand what the impact of various things, including human activity, will be over decades or centuries.

Both climatology and meteorology are branches of physical science that deal with the weather. While they are related to one another in many ways, they aren't the same thing. As the saying goes, the weather changes frequently, but the climate is relatively stable. That saying defines the major difference between the two sciences, which is basically one of time.

History of Climatology

Early references to the weather can be found in the poems of ancient Greece and in the Old Testament of the Judaeo-

Christian Bible. Even older references can be singled out in the Vedas, the most ancient Hindu scriptures that were written about 1800 BCE. Specific writings on the theme of meteorology and climatology are found in Hippocrates' Air, Waters, and Places dated around 400 BCE followed by Aristotle's Meteorologica written around 350 BCE. To the early Greek philosophers, climate meant "slope" and referred to the curvature of the Earth's surface which, due to the changing incidence of the sun's rays, gives rise to the variation of climate with latitude. Logical and reliable inferences on climate can be found in the works of the Alexandrian philosophers Eratosthenes and Aristarchus.

With the onset of extensive geographical exploration in the 15th century, descriptions of the earth's climates and the conditions giving rise to the climates started to emerge. The invention of meteorological instruments such as the thermometer in 1593 by Galileo Galilei and the barometer in 1643 by Evangelista Torricelli gave a greater impulse to the establishment of mathematical and physical relationships between the different characteristics of the atmosphere and, therefore, of relationships which could describe the state of the climate at different times and in different places.

The observed pattern of circulation linking the tropics and subtropics, including the trade winds, tropical convection, and subtropical deserts, was first interpreted by George Hadley in 1735, and thereafter known as the Hadley Cell. Julius von Hann, who published the first of three volumes of the Handbook of Climatology in 1883, wrote the classic work

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on general and regional climatology that included data and eyewitness descriptions of weather and climate.

In 1918 Wladimir Köppen produced the first detailed classification of world climates, based on the vegetative cover of land. This endeavor was followed by more detailed development in descriptive climatology. The geographer E.E. Federov, for example, attempted to describe local climates in terms of daily weather observations.

In the first thirty years of the 20th Century the diligent and combined use of global observations and mathematical theory to describe the atmosphere, led to the identification of large scale atmospheric patterns. Notable in this field was Sir Gilbert Walker, who conducted detailed studies of the Indian monsoon, the Southern Oscillation, the North Atlantic Oscillation, and the North Pacific Oscillation.

Other major works on climatology included those by Tor Bergeron (on dynamic climatology in 1928) and Wladimir Köppen and Rudolf Geiger (a climatology handbook in 1936). Geiger first described the concept of microclimatology in some detail in 1927, but the development of this field did not occur until the Second World War. During the war, for planning purposes, a probability risk concept of weather data for months or even years ahead was found to be necessary and was tried. C.W. Thornthwaite established a climate classification in 1948 based on a water budget and evapotranspiration. In the following decades development of theories on climatology saw major progress.

During the latter decades of the 20th century the issue of climate change began to focus attention on the need to understand climate as a major part of a global system of interacting processes involving all of the earth's major domains. Climate change is defined as a statistically significant variation in either the average state of the climate or in its variability, persisting for an extended period, typically decades or longer. It may be caused by either natural internal processes, external forcing, or by persistent anthropogenic (resulting from or produced by human activity) changes in the composition of the atmosphere or in land use.

Climatology Investigation Methods

There are different approaches in Climatological investigations.

- Paleoclimatology which seeks to reconstruct past climates through examining records such as ice cores and tree rings (dendroclimatology);
- Use of meteorological data accumulated over many years to study contemporary climates;
- Use of statistical or mathematical models to study the atmosphere and its dynamics;
- Modeling for understanding past, present and potential future climates; and

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- Historical climatology where climates are related to human history and thus focuses only on the last few thousand years.

Properties

Three properties of the climate include "normals," "extremes," and "frequencies" and are used to gauge the state of the atmosphere over a particular time period

- "Normals" refers to average weather conditions at a place
- "extremes" are used to describe the maximum and minimum measurements of atmospheric variables
- "frequencies" refers to the rate of incidence of a particular phenomenon at a particular place, over a long period of time

Since climate is a long term representation of weather on a large scale and is a product of complex processes, researchers need to have very good understanding of physical laws usually expressed as differential equations. This makes it difficult to conduct climate research since the

equations are normally coupled and are non-linear, thus requiring numerical methods to obtain approximate solutions. Sometimes climate is modeled as a stochastic process when approximating very complex processes. These activities necessitate the quantitative approach in climatology.

The Climate System

The climate system is a complex, interactive system consisting of the atmosphere, land surface, snow and ice, oceans and other bodies of water, and living organisms. The atmosphere is the gaseous envelope surrounding the Earth. The dry atmosphere consists almost entirely of nitrogen and oxygen, but also contains small quantities of argon, helium, carbon dioxide, ozone, methane, and many other trace gases. The atmosphere also contains water vapor, condensed water droplets in the form of clouds, and aerosols.

The hydrosphere is that part of the Earth's climate system comprising liquid water distributed on and beneath the Earth's surface in oceans, seas, rivers, freshwater lakes, underground reservoirs, and other water bodies. The cryosphere collectively describes elements of the Earth system containing water in its frozen state and includes all snow and ice (sea ice, lake and river ice, snow cover, solid precipitation, glaciers, ice caps, ice sheets, permafrost, and seasonally frozen ground). The surface lithosphere is the upper layer of the solid Earth, both continental and oceanic. The biosphere comprises all ecosystems and living organisms in the atmosphere, on land (terrestrial biosphere), or in the oceans (marine biosphere), including derived dead organic matter, such as litter, soil organic matter, and oceanic detritus.

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The climate system

Under the effects of solar radiation and the radiative properties of the surface, the climate of the Earth is determined by interactions among the components of the climate system. The interaction of the atmosphere with the other components plays a dominant role in forming the climate. The atmosphere obtains energy directly from incident solar radiation or indirectly via processes involving the Earth's surface. This energy is continuously redistributed vertically and horizontally through thermodynamic processes or large scale motions with the unattainable aim of achieving a stable and balanced state of the system. Water vapor plays a significant role in the vertical redistribution of heat by condensation and latent heat transport. The ocean, with its vast heat capacity, limits the rate of temperature change in the atmosphere and supplies water vapor and sensible heat to the atmosphere. The distribution of the continents affects oceanic currents, and mountains redirect atmospheric motions. The polar, mountain, and sea ice reflect solar radiation back to space. In high latitudes the sea ice acts as an insulator and protects the ocean from rapid energy loss to the much colder atmosphere. The biosphere, including its human activities, affects atmospheric components such as carbon dioxide as well as features of the Earth's

surface such as soil moisture and albedo.

Interactions among the components occur on all scales Spatially, the microscale encompasses features of climate characteristics over small areas such as individual buildings and plants or fields. A change in microclimate can be of major

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importance when the physical characteristics of an area change. New buildings may produce extra windiness, reduced ventilation, excessive runoff of rainwater, and increased pollution and heat. Natural variations in microclimate, such as those related to shelter and exposure, sunshine and shade, are also important, for example, from determining which plants will prosper in a particular location, to the need to make provision for safe operational work and leisure activities.

The mesoscale encompasses the climate of a region of limited extent, such as a river catchment area, valley, conurbation, or forest. Mesoscale variations are important in applications including land use, irrigation and damming, the location of natural energy facilities, and resort location.

The macroscale encompasses the climate of large geographical areas, continents, and the globe. It determines national resources and constraints in agricultural production and water management, and is thus linked to the nature and scope of human health and welfare. It also defines and determines the impact of major features of the global circulation such as the El Niño Southern Oscillation (ENSO), the monsoons, and the North Atlantic Oscillation.

A temporal scale is an interval of time. It can range from minutes and hours to decades, centuries, and longer. The characteristics of an element over an hour are important, for example, in agricultural operations such as pesticide control and in monitoring energy usage for heating and cooling. The characteristics of an element over a day might determine, for example, the human activities that be safely pursued. The climate over months or years will determine, for example, the crops that can be grown or the availability of drinking water and food. Longer time scales of decades and centuries are important for studies of climate variation caused by natural phenomena such as atmospheric and oceanic circulation changes and by the activities of humans.

Contributions of Climatology

Climatology has become a dynamic and highly functional branch of science. New techniques are being developed and investigations undertaken to study the application of climate in many fields such as agriculture, forestry, ecosystems, energy, industry, production and distribution of goods, engineering design and construction, human well-being, transportation, tourism, insurance, water resource and disaster management, fisheries, and coastal development.

The crucial roles of climate data and climate predictions in planning for disaster mitigation and sustainable development, and in addressing all the consequences of climate change are now firmly established within various conventions such as the United Nations Framework Convention on Climate Change.

Applied climatology makes the maximum use of meteorological and climatological

knowledge and information for solving practical social, economic, and environmental problems. Climatological services are designed for a variety of public, commercial, and industrial users. Assessments of the effects of climate variability and climate change on

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human activities, as well as the effects of human activities on climate, are major factors in local, national and global economic development, social programmes, and resource management.

Current interest on the consequences of economic and human activities on climate and how climate variability and change influence human societies highlights the need for further research into the physical and dynamical processes in the climate system and for their statistical description. Understanding of natural climate variability, appreciation of climate sensitivity to human activities, and insight on the predictability of the weather and climate for periods ranging from days to decades is fundamental to improving our capability to respond effectively to economic and societal problems.

Precisely, fundamental importance of climatology can be highlighted as follows:

- Climatology seeks to explain the causes of different types of climates
 - o The reasons for their variations
 - o Their effects on natural vegetation
 - o The processes that produce different climates
- Climatology makes a detail analysis of the interaction of weather and climatic elements upon human societies
- Climatology discusses the various climatic elements, the factors that control the distribution of climate over the Earth
- Climatology is important in determining the climatic patterns of a particular region. Establishing the climatic pattern is significant in deciding the economic activities that would thrive in that particular region. If the climate of a region is established to be cool and wet, it would be safe to conclude that agriculture might thrive in the region.
- Having a clear climate pattern makes it easier for people to understand the seasons of engaging in particular tasks. This is especially most important to tourists and farmers.
- Infrastructure development, especially buildings are dependent on climate. After a climatic pattern has been established, the engineers recommend the use of materials that would not only withstand the conditions but also protect the dwellers from any harsh climatic condition.

Contemporary Climatological Challenges

Climate change has become a major issue for the human community. Human activities, especially the burning of fossil fuels, have led to changes in the composition of the global atmosphere. Marked increases in tropospheric carbon dioxide and methane during the industrial era, along with increased aerosols and particulate emissions are significantly affecting global climate. Chlorofluorocarbons extensively used in the past as aerosol propellants, cleaning fluids, and refrigerants are a main cause of stratospheric ozone

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depletion. Over one fifth of the world's tropical forests were cleared between 1960 and 2000, likely altering the complex mesoscale and global hydrological cycles. Artificial canyons formed in cities by buildings, together with the asphalt of roads, increase the amount of radiation absorbed from the sun and form urban heat islands. Accelerated runoff of rainwater and the removal of trees and other vegetation reduce the amount of transpired water vapor that would otherwise help to moderate temperature. Pollution from vehicles and buildings accumulates, especially in calm conditions, and causes many human health problems and damage to structures.

Introductory Biogeography

Definitions

The term Biogeography is derived from the union of two words "Bio" and "Geography". Bio refers to biology, a science that deals with living phenomena while geography is a science that is concerned with distribution and interrelationships between both living and nonliving phenomena. Biogeography as a branch of physical geography is the study of spatial distribution of both plants and animals and the processes that produce variations in the patterns of distribution (Browne, 1993). Biogeography is considered as an interdisciplinary branch of geography and biology which deals with the spatial and temporal patterns of flora and fauna in respect of surviving geographical conditions.

Biogeography refers to the study of distribution of species and ecosystems in geographic space through geological time. Biogeography is often studied in the context of ecological and historical factors which have shaped the geographical distribution of organisms over time. Specifically, species vary geographically based on latitude, habitat, segregation/isolation (e.g., islands), and elevation.

Biogeography implies a linkage between Biology and Geography. It studies the distribution of biological materials over the earth's surface and the factors responsible for the observed spatial variations (Pears, 1977). Jones (1980) described Biogeography as a subject that studies the expansion pattern of organisms in the past and present.

Biogeography is the study of past and present-day spatial distributions of life (both flora and fauna) on earth in relation to environmental factors such as geology, climate and soils.

Nature and Scope

Biogeography enables us to identify biodiversity patterns in the past and present, identify the expansion of organisms which enables us to study about the relationship between living and non-living factors that influence organisms to exist. Diversity of flora and fauna exists according to the relationship that emerges with these factors.

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The field of biogeography covers the trio-spheres of the earth viz- hydrosphere, lithosphere and atmosphere. Combination of these three gives rise to a fourth sphere known as biosphere. More precisely it is the study of the biologically inhabited part of the Earth therefore known as

Biosphere (De Quiroz, 2005). Its subject matter covers the multitudinous forms of plant and animal life which inhabit „biosphere“ a relatively shallow but densely populated zone as well as the complex biological processes which control their activities.

Biogeography is an interdisciplinary natural science with a complex, even cryptic identity that defies definition. In essence, it is the science of the distribution of living organism and the factors that underlie this distribution. Not only is biogeography concerned with spatial patterns in the contemporary sense, but also there is a strong evolutionary component and a concern with change over time (Meadows, 2015).

Within biogeography there are various degrees of specialization, usually involving taxonomic study of either fauna (zoogeography) or flora (phytogeography). As such, biogeography depends heavily upon the observations and concepts of its parent subjects– geography, earth science and biology, and especially ecology and paleoecology (Pitman, 2011).

Summarily, nature and scope of biogeography can be outlined as:

- Biogeography studies the cooperative relationship between vegetation without intelligence and the man with intellectual capacity.
- Biogeography mainly pays its attention towards the diversities according to spatial changes.
- Biogeography studies different changes of animal life
- Biogeography studies how different types of animals get acclimatized to the environment. Changes in flora and fauna life take place according to the climate. As for animals, animals living in warm regions are different from animals living in cold regions.
- Biogeography studies the lithosphere; living beings of interior earth, there living begins, soil and rocks. It studies more on soil than rocks and also about mass movement of micro ecological systems.
- Attention is paid towards human effects in addition to physical and biological factors. Especially, the natural flora changes in agricultural process. (Eg: clearing of forests for cultivation, changes the ecological condition which existed before). Accordingly, biogeography studies about the human effects towards the environment.

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Branches

There are several branches/fields of biogeography and each addresses the distribution of species from a different perspective.

1. Phytogeography is the branch of biogeography that studies the distribution of plants.
2. Zoogeography is the branch that studies distribution of animals.
3. Historical biogeography primarily involves specie distributions from an evolutionary perspective. Studies of historical biogeography involve the investigation of phylogenic distributions over time.
4. Ecological biogeography refers to the study of the contributing factors for the global distribution of plant and animal species. Some examples of ecological factors that are commonly studied include climate, habitat and primary productivity (the rate at which the

plants in a particular ecosystem produce the net chemical energy). Moreover, ecological biogeography differs from historical biogeography in that it involves the short-term distribution of various organisms, rather than the long-term changes over evolutionary periods.

5. Conservation biogeography seeks to effectively manage the current level of biodiversity throughout the world by providing policymakers with data and potential concerns regarding conservation biology.

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Investigation Approaches

The major approaches in biogeography are classified broadly into those rooted in the spatial tradition and those associated with a more ecological theme, as shown in the schematic representation below (Meadows, 2015).

Spatial Tradition: Focus is on Phytogeography, Zoogeography, Historical Biogeography and Vicariance dispersal. Phytogeography and zoogeography are concerned with the geographic distribution of the plant species and animal species respectively. Historical biogeography study about how species distribution have changed overtime in relationship to the history of landforms, ocean basis and climate as well as how those changes have contributed to the evolution of biota. Vicariance dispersal means how species have been distributed and the division of a widespread group of organisms by a geographic barrier, such as a mountain range or a body of water, often resulting in the evolution of related species on either side of the barrier.

Ecological Tradition: Focus is on study of Ecosystem, Paleoecology and Island Biogeography. Ecology is the study of the interaction among organism and an ecosystem is comprised of habitats, biological communities, and ecotones (a transition area between two biomes or where two communities meet). Therefore to understand the distribution of species on earth a fundamental knowledge of ecology and ecosystems is essential. Paleoecology study fossil of animals and plants in order to realize their ecology and the environmental conditions in which

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they lived. Island biogeography seek to understand the generation of biodiversity on island archipelagos in relation to geographical isolation and dispersal abilities of taxa.

Comparative Biogeography: The study of comparative biogeography can follow two main lines of investigation; Systematic biogeography, the study of biotic area relationships, their distribution, and hierarchical classification and Evolutionary biogeography, the proposal of evolutionary mechanisms responsible for organismal distributions. Possible mechanisms include widespread taxa disrupted by continental break-up or individual episodes of long-distance movement.

In spite of the fact that this field of study is a transition between biology and geography; it is not the exclusive preserve of either of these two sciences. By its very character biogeography is situated at and overlaps the boundaries of a great number of other disciplines. Disciplines like geology, climatology, pedology, geomorphology as well as the botany, zoology, genetics and geography contribute to this very large and varied field. In doing so they are to a greater extent,

dependent on an understanding of biogeography. As a result, the approach to biogeography in large measure is determined by the interest and objectives of the particular scholar of a specific scientific field (Lomolino, 2000).

Thrust of Inquiry

The basic material for biogeography investigation consists of the organisms, plant and animal communities which exhibit wide extension over the globe with varying degrees of the biosphere which envelops the surface of the earth and consists of all living-organisms living on land, aquatic and aerial areas.

The patterns of species distribution across geographical areas can usually be explained through a combination of historical factors such as: speciation, extinction, continental drift, and glaciation. Through observing the geographic distribution of species, we can see associated variations in sea level, river routes, habitat, and river capture. Additionally, this science considers the geographic constraints of landmass areas and isolation, as well as the available ecosystem energy supplies. Biogeography explores questions on whether the situation of identifying the life styles that existed in the past are being practiced in the present, whether those practices are being followed in past have been subjected to changes and what the factors that influenced such changes. It is important to point out that in both biological and geography literature attention has been given to primarily though not exclusively concerned with the study of plant rather than animal geography. There are various reasons for this. Greater mobility combined with the small size and an elusive habit of life of the majority of animal species make a study of their distribution more difficult than that of plants. Also, until fairly recent, zoogeography had not developed to anything like the same extent as plant geography. Other considerations have however, been responsible for the geographer"s preoccupation with plant life. Plants-vegetable matter-both living and decaying, comprise the greatest bulk of the total world biomass both above and below

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the ground surface and in water bodies. Hence plants are the most conspicuous components of the biosphere and en masse major landscape element. Further, plants are more directly depend upon their physical habitat than animals. In comparison to the latter, plants are relatively immobile or stable due to lack of locomotive organs. As a result they provide a better index of the total environment -physical, biological and human. Because of their greater biomass, plants exert a greater influence on the character of the atmosphere and soil they occupy. They not only modify the physical habitat, but in doing so they create a particular biological environment which would not otherwise exist. Finally plants provide the primary source of food -energy for all other living organisms-including man. They are basic to the geography of animals and constitute the most important of man"s resources (Holden et. al., 2004).

Also, biogeography is most keenly observed on the world's islands. These habitats are often much more manageable areas of study because they are more condensed than larger ecosystems on the mainland. Islands are also ideal locations because they allow scientists to look at habitats that new invasive species have only recently colonized and can observe how they disperse throughout the island and change it. They can then apply their understanding to similar but more complex mainland habitats. Islands are very diverse in their biomes, ranging from the tropical to

arctic climates. This diversity in habitat allows for a wide range of species study in different parts of the world.

Modern biogeography often employs the use of Geographic Information Systems (GIS), to understand the factors affecting organism distribution, and to predict future trends in organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

As such, in terms of applications of biogeography as a science today, technological advances have allowed satellite imaging and processing of the Earth. Two main types of satellite imaging that are important within modern biogeography are Global Production Efficiency Model (GLO-PEM) and Geographic Information Systems (GIS). GLO-PEM uses satellite-imaging gives "repetitive, spatially contiguous, and time specific observations of vegetation". These observations are on a global scale. GIS show certain processes on the earth's surface e.g. whale locations, sea surface temperatures, and bathymetry (Underwater – ocean, sea, lakes, rivers etc. - depths and topography) etc. Current scientists also use coral reefs to delve into the history of biogeography through the fossilized reefs.

Basic Concepts

Some fundamental concepts in biogeography include:

□ **Species** – A species is a group of organisms that share a genetic heritage, are able to interbreed, and to create offspring that are also fertile. A species is often defined as the largest group of organisms in which two individuals can produce fertile offspring, typically by sexual reproduction. A grouping of the same type of organisms.

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□ **Populations** – A group of species in a specific area at a specific time. Population is all the organisms of the same group or species, which live in a particular geographical area, and have the capability of interbreeding. A group of individuals of the same species occupying a particular geographic area.

□ **Communities** – All populations in a specific area at a specific time. In ecology, a community is a group or association of populations of two or more different species occupying the same geographical area and in a particular time.

□ **Ecosystems** – All living and non-living things in their environment as well as their interactions. An ecosystem includes all of the living things (plants, animals and organisms) in a given area, interacting with each other, and also with their non-living environments (weather, earth, sun, soil, climate, atmosphere) - food chain, food web, nutrient cycles

□ **Ecozones** - An ecozone (or biogeographic realm) is the broadest biogeographic division of the Earth's land (or water) surface, based on distributional patterns of terrestrial (or aquatic) organisms. They are subdivided in ecoregions, which are classified in biomes or habitat types. Each ecozone may include a number of different biomes. E.g. terrestrial ecozones include Nearctic, Afrotropics and Oceania realms, marine include eastern pacific, western Atlantic regions etc.

□ **Ecoregions** – Large unit of land or water containing a geographically distinct assemblage

of species, natural communities and environmental conditions. They are a division of the larger "ecozones" or biogeographic regions. E.g. desert, forest (mangrove, rainforest), grassland (guinea, sudan, sahel) etc.

□ Ecological succession – The non-seasonal change of plant communities over time.

Primary (e.g. volcanic explosion) & Secondary (hurricane). Pioneer species (first plant species to return to an area during primary succession)

□ Allopatric Speciation – This is the splitting of a species by evolution of geographically isolated populations

□ Evolution – change in genetic composition of a population

□ Extinction – disappearance of a species

□ Dispersal – movement of populations away from their point of origin, related to migration

□ Endemic Areas – Locations or populations where a specie exists permanently

□ Geodispersal – The erosion of barriers to biotic dispersal and gene flow, that permit range expansion and the merging of previously isolated biotas

□ Range and Distribution - The range of a species is the geographical area within which that species can be found. Within that range, distribution is the general structure of the species population

□ Vicariance –It is the separation or division of a group of organisms by a geographic barrier, such as a mountain or a body of water, resulting in differentiation of the original group into new varieties or species. This can also be referred to as fragmentation of the

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environment or geographical separation typically by a physical barrier (such as a mountain range, river or splitting of a tectonic plate) in contrast to dispersal as a factor in promoting biological evolution by division of large populations into isolated subpopulations.

Link between Biogeography and Climatology

Climatology deals with study of various elements and interrelated weather phenomena which affects life on the earth. Under normal circumstances climate is essential to the growth of plants and animals including human beings but in adverse circumstances like drought and flood it disturbs the lives and their habitats. Therefore climate is an inseparable part of biogeography. Its influence can be seen in some biogeographical themes such as Agriculture, Physiological adaptation, Dispersal, Range etc.

Applied climatology studies the application of climatic knowledge to solve various problems faced by human society e.g. global warming and climate changes. It studies the interactions between climate and biosphere i.e. how does climate influence and control plants and animals including man and in turn how does man modify climate by introducing advertent and inadvertent changes in the physical environment and by making certain weather modification e.g. cloud seeding to induce precipitation.

Generally, atmospheric processes particularly the climate significantly influences the biotic and

abiotic components of the terrestrial and aquatic ecosystems. Therefore, problem solving in environmental issues has to adopt both interdisciplinary and multidisciplinary approach.

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